

Farbod Raeisi

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Education

B.Sc. Electrical Engineering | 2021 - 2025

K.N. Toosi University of Technology, Tehran, Iran
GPA: 3.5 of 4 (17.01/20) via 133 credits

High School Diploma (Mathematics and Physics) | 2017 - 2021

Allame Helli High School, National Organization for Development of Exceptional Talents (NODET)
GPA: 18.69 of 20

Publications

- F. Raeisi, Dr. M. Delrobaei, "Design and Implementation of an Intelligent Vision-Sensor System for Monitoring Mind-Wandering Level through Facial Movement Analysis" (Bachelor's Thesis, to be submitted to conference)

Research Experience

Mechatronic Laboratory

September 2023 – *present*

Researcher

- Under supervision of [Dr. Mehdi Delrobaei](#)
Working on a research paper about assessment of mind wandering using visual signals.

SBMU Neuroscience Research Center (NRC)

August 2024 – *January 2025*

Research Assistant

- Under supervision of [Dr. Mahdi Aliyari-Shorehdeli](#)
Assisting in addressing LFP data acquisition challenges from rat brains to support neuroscience research initiatives.

Fault Detection & Identification Laboratory (FDI)

August 2023 – March 2024

Research Assistant

- Under supervision of Dr. Mahdi Aliyari-Shorehdeli
Part of a team analyzing NHTS data for a master's project, aimed at refining airbag performance prediction systems by evaluating American car crash sensor data.

Internships and Summer Schools

Neuromatch Academy | Computational Neuroscience Program

Summer 2023

- Gained hands-on experience in computational neuroscience, analyzing the Steinmetz dataset.
Collaborated with an international team on data analysis techniques

Machine Learning and Image Processing Intern | Fard Iran Inc.

Summer 2022

- Contributed to developing a car plate recognition system, using machine learning algorithms and image processing to improve accuracy.

Teaching Experience

Linear Control Course

September 2024 – February 2025

[Prof. Hamidreza Taghirad](#) | K. N. Toosi University of Technology

Advanced Programming Course

September 2024 – January 2025

[Hossein Yekta Moghadam](#) | University of Tehran

Probability and Statics Course

October 2022 – July 2023

[Dr. Bahare Akhbari](#) | K. N. Toosi University of Technology

Selected Projects

Control and Simulation of a Robotic Arm Utilizing LQR Optimization

- Implemented LQR optimization to design an efficient control system for a robotic arm. The system's dynamics were modeled in SolidWorks, and simulations were run in MATLAB Simulink to test and refine performance.

Design and Implementation of State Controller for Propellant Spacecraft System

- Conducted a detailed analysis of propellant spacecraft systems, deriving state equations, evaluating controllability and observability, and implementing state feedback and LQR control strategies.

Assessing Choice Certainty as a Predictor of Performance Accuracy in Mice Using the Steinmetz Dataset

- Focused on decoding neural signals to explore how certainty in decision-making correlates with performance accuracy across different brain regions. This experience enhanced my understanding of computational neuroscience and research methodologies.

Load Flow Analysis and Contingency Assessment of a 13-Bus Power System Using DIgSILENT

- Used DIgSILENT software to conduct AC and DC load flow analyses on a 13-bus power system, evaluating the impact of contingencies like line and generator outages. The analysis offered valuable insights into system performance and reactive power constraints under various scenarios.

Automated Vending Machine Simulation Through Digital Circuit Design Techniques Using Proteus

- Designed and simulated an automated vending machine for two chocolate types using Proteus. Developed a state table, state diagram, and optimized the circuit with flip-flops to handle 1-cent, 2-cent, and 4-cent coins, ensuring accurate display of the entered amount and proper functionality.

Skills

Artificial Intelligence

- Machine Learning
- Reinforcement Learning
- Deep Learning
 - Convolutional Neural Networks

Software

- SolidWorks
- Unreal Engine 5
- Altium Design
- Arduino IDE
- Raspberry Pi OS
- Proteus
- SPSS
- PSpice
- COMSOL Multiphysics
- CodeVision AVR

Programming Languages

- Python
- PyTorch
- C/C++
- html / CSS
- SQL

Microcontrollers

- ESP32 Microcontrollers
- Raspberry Pi

MATLAB

- System identification toolbox,
- Neural network fitting
- Simscape
- Simulink
- Fuzzy Logic Toolbox
- PID Tuner App

Award & Honors

- Ranked 15th (top 10%) out of 150 students in the K. N. Toosi University of Technology, based on GPA and 3rd in control focused students
- Admitted to National Organization for Development of Exceptional Talent

Volunteer Experience

- **Executive Committee Member** | The 3rd International Conference on Electrical Machines and Drives (ICEMD 2023)
- **Executive Committee Member** | The 6th International Conference on Millimeter Wave Terahertz Technologies (MMWATT 2023)
- **Photographer** | IEEE Cultural Branch Iran Section member at K.N. Toosi University of Technology
- **Mentor** | Open Doors Day at Mechatronics Lab
- **Fundraising Participant** | Fundraiser for Flood Relief Victims in Western Iran 2017

Language Proficiency

- English: Full professional proficiency
Academic IELTS band score 7.5 (Listening: 8.0 Reading: 8.5 Speaking: 7.5 Writing: 6)
- German: A2 level
- Persian: Native

Courses & Certificate

Python for Data Science and Machine Learning Bootcamp • Udemy	August 2024
Signal Processing Problems, solved in MATLAB and in Python • Udemy	August 2024
Applied Electronics for Robotics • ARAS (Advanced Robotics and Automated System) Hi-Tech Robotic Solutions Group	June 2023
Introduction to the Internet of Things and Embedded Systems • Coursera	August 2023
Computational Neuroscience • Neuromatch International Academy	September 2023

References

Available upon Request